



Optimizing Device Financing Decisions Through Predictive Risk Modeling

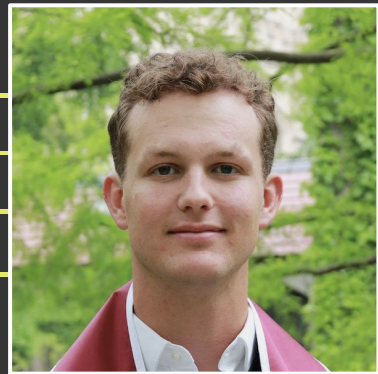
The Outliers



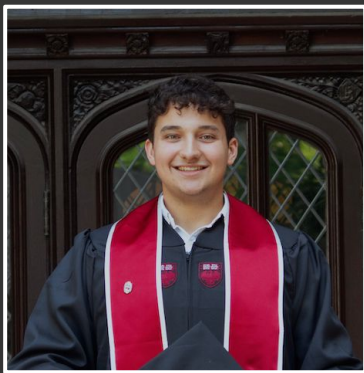
Meet the Team



Sola Shin
Finance Manager



Kirk Waller
Data Analyst



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Technical Lead



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Business Analyst



Szuyu Chi
Franchise Manager



Xander Deanhardt
Data Scientist

Overview



The Issue

Verizon currently lacks a better method of **differentiating high-risk from low-risk** financing applicants, leading to **unnecessary default losses** from delinquent customers and **missed revenue** from customers who would have otherwise paid.

What Does This Mean for You?

Goals

- ❑ Approve more **good, paying customers**
- ❑ Reduce **device financing losses**
- ❑ Make **faster, consistent** in-store decisions
- ❑ Improve profitability of device financing

What's Going Wrong Today

- ❑ Current approval rules are **too broad** → many risky customers approved
- ❑ **False negatives** → rejecting customers who would pay (lost LTV)
- ❑ **False positives** → approving customers who default (lost device cost)

You need a data-driven in-store approval model that reduces default losses AND protects valuable paying customers.

What This Means for Verizon

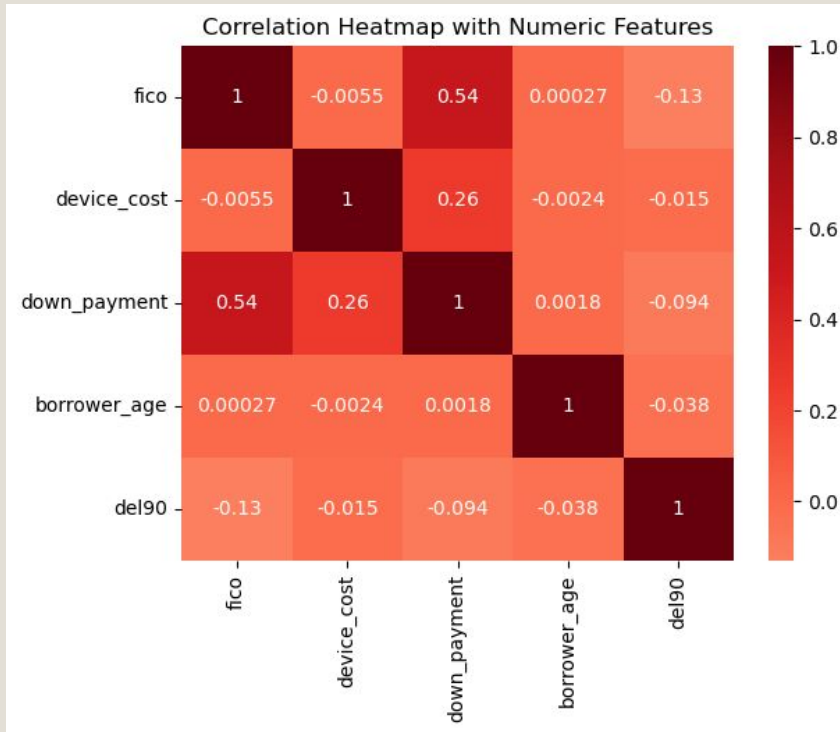
Less false negatives
=
More paying
accounts being
opened

- At a **net LTV of \$4,780 per account¹**, an **average defaulting cost of \$811**, and a **postpaid churn rate of only 0.91%²**, every misclassified customer has permanent financial consequences: good customers who are rejected *rarely return*, and bad customers who are approved create *avoidable losses*.

Less false positives
=
Fewer delinquent
accounts being
opened

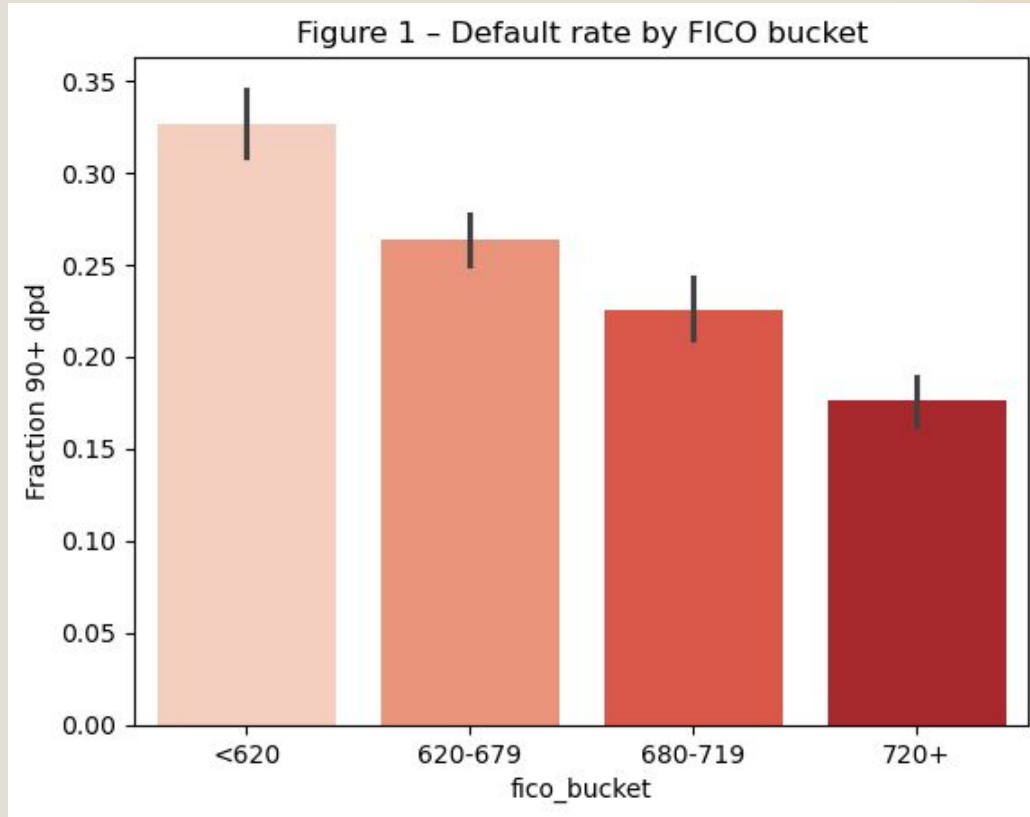
- A false-negative costs nearly 6x more than a false-positive**, and losing a single good customer risks the entire household bundle
- Verizon added 306,000 broadband customers in Q3 alone³ and more households entering the ecosystem means more device-financing decisions where accuracy matters.

Our Insights



- del90 - Target variable describing whether borrower was ever 90+ days late with payment or defaulted on loan
- fico - Borrower's FICO credit score
- device_cost - Price of the device being financed in USD
- down_payment - Initial payment made by the borrower in USD
- borrower_age - Borrower age in years

Our Insights



Modeling Approach

verizon

Age_t: sqrt (borrower_age)
Coef: 0.0993

Fico_t

sqrt

$(fico - (fico - 350) + (800 - fico))$

Fico_t x
Tier 4

Coef: -.24

Fico_t x
Tier 3

Coef:
-.20

Fico_t x
Tier 2

Coef: -.34

Fico_t x
Tier 1

Coef:
-.45



Tier 4:
\$5.2-\$32



Tier 3:
\$32.1-\$36.7



Tier 2:
\$36.8-\$39.5



Tier 1:
\$39.5-\$60

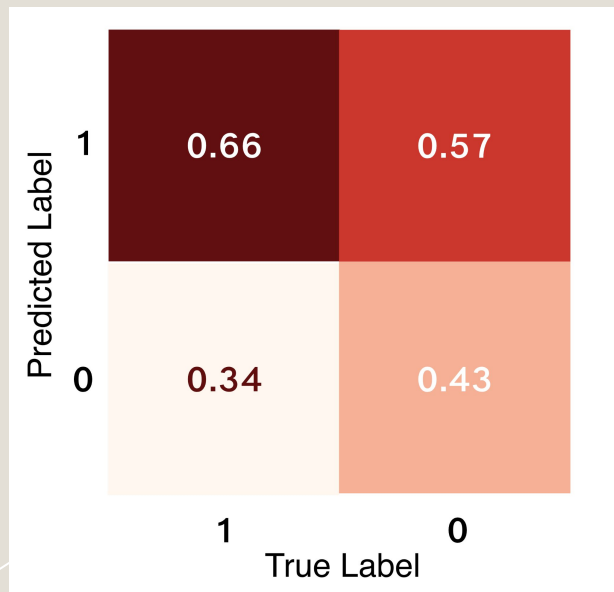


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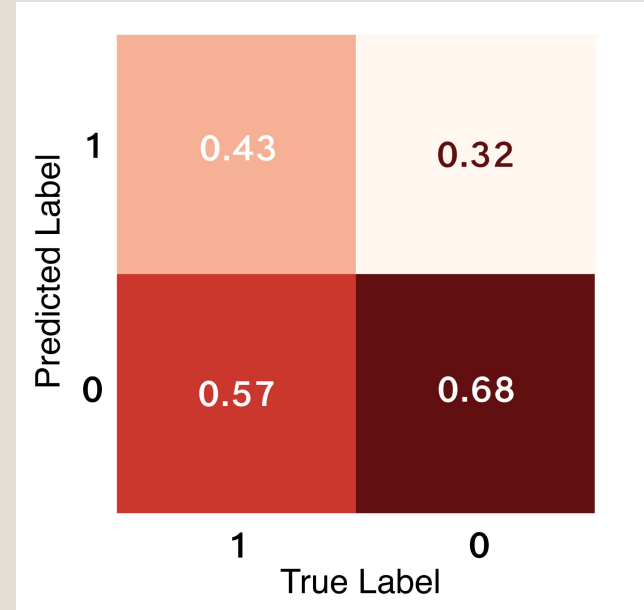
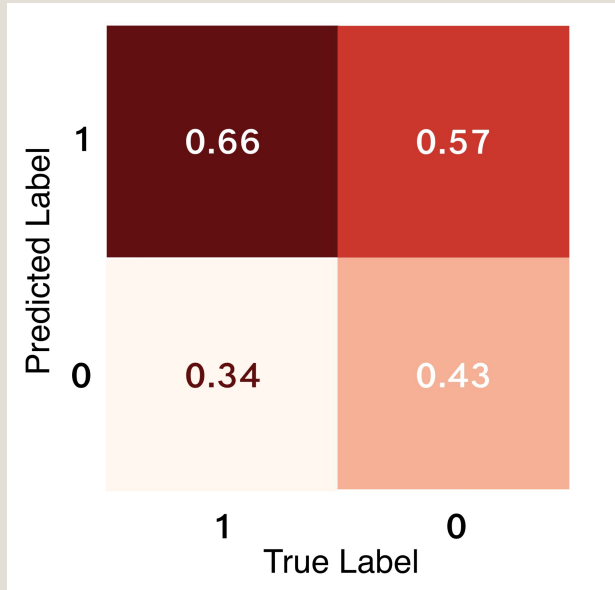
Model Performance

- Verizon model's True Positive Rate (TPR): **0.429**
 - Assumed model was Logistic Regression using 'fico' as only predictor
- Team Outlier's model's TPR: **0.661**
- Major improvements with new features
 - **54% increase in predictive power**

Our Model Outcome:



Our Model vs Verizon's Model



Business Value Impact (1M customers)

What changes with our model?

- F1 score : 0.57 -> 0.722
- Precision : 0.849 -> 0.796
- Recall : 0.429 -> 0.661
- Accuracy : 0.476 -> 0.608

Headline Result

- Net Business Value(29.4% Margin) : **+\$1.65B**

Key Inputs

- Loss per default : \$811 (Avg Remaining Balance, Defaults Only)
- LTV : (ARPA * Margin Rate) / Monthly churn
Gross: \$16,259
At 29.4% Margin : \$4,780

Business Value Impact_(1M customers)

Profit / Loss by Category

Category	Definition	Unit Price	BaseLine (FICO only)	New Model (Age + FICO Interactions)	Δ Revenue from New Model
TP	Correctly approved good customers → earned LTV	+\$4,780	+\$1,659,722,222	+\$2,432,417,806	+\$772,695,584
TN	Correctly rejected risky customers → avoided loss	+ \$811	+\$104,625,000	+\$80,599,359	-\$24,025,640
FN	Wrongly rejected good customers → missed LTV	- \$4,780	-\$2,208,094,444	-\$1,248,891,475	+\$959,202,969
FP	Wrongly approved risky customers → realized loss	- \$811	-\$49,950,000	-\$105,580,448	-\$55,630,448

Net Value Formula

$$\text{Net} = (\text{TP} \times \text{LTV}) + (\text{TN} \times L_{\text{Avoided}}) - (\text{FN} \times \text{LTV}) - (\text{FP} \times L_{\text{Loss}})$$

Net Increase in Business Value

+ 1.65 B

Policy Recommendations

- ❑ Even though the model demonstrates large potential financial upside, it must be validated in real operational settings which would necessitate a phased rollout

Shadow

Existing Verizon model remains active with the new model running in parallel behind the curtains, not affecting approval/rejection decisions to determine performance

Test

implement in a percent of stores with “stable” customer bases and in a percent stores with “riskier” customers bases to establish in-use performance

Fine-Tune

Using collected data from pilot implementation, fine-tune the new model and rerun pilot-testing and fine-tuning as needed

Rollout

Begin incrementally adopting model at more and more stores nationwide until full adoption

Next Step with Verizon

We require the following key parameters from Verizon to precisely calculate the Business Value of our solution, addressing both risk reduction and opportunity cost.

Area	Data Needed	Why it Matters
Default Criteria & Rate	Official 90+ day delinquency standards.	Validates the model's risk threshold and ensures our loss simulation aligns with Verizon's internal definitions.
True Net Loss per Default	Average remaining principal at default & Recovery Rate (e.g., 85% Balance / 5% Recovery).	Determines the actual cash loss for every False Positive, allowing us to calculate the value of correctly rejecting bad customers .
Device Financing Margin	Gross / Operating Margin by product line.	Converts gross revenue to Net Lifetime Value (LTV) . This is crucial for calculating the Opportunity Cost of rejecting a good customer .
ARPA & Churn Rate	Average monthly revenue and churn rate.	Core inputs for the LTV formula, providing the projected revenue stream from a correctly approved customer.
Portfolio Composition	% of financed customers and FICO mix.	Scales our per-million applicant estimates to the entire enterprise level, validating the scope of the solution.

Preliminary Plan

Week 1-2

Project Initiation &
Data Preparation
Scope Finalization &
Data Quality
Assurance.

Week 3-4

Exploratory Analysis
& Feature Selection
Interpretable Risk
Predictors (FICO,
D/P Ratio).

Week 5-8

Model Development
& Evaluation
High-Performance
XGBoost Model &
Stability Testing.

Week 9

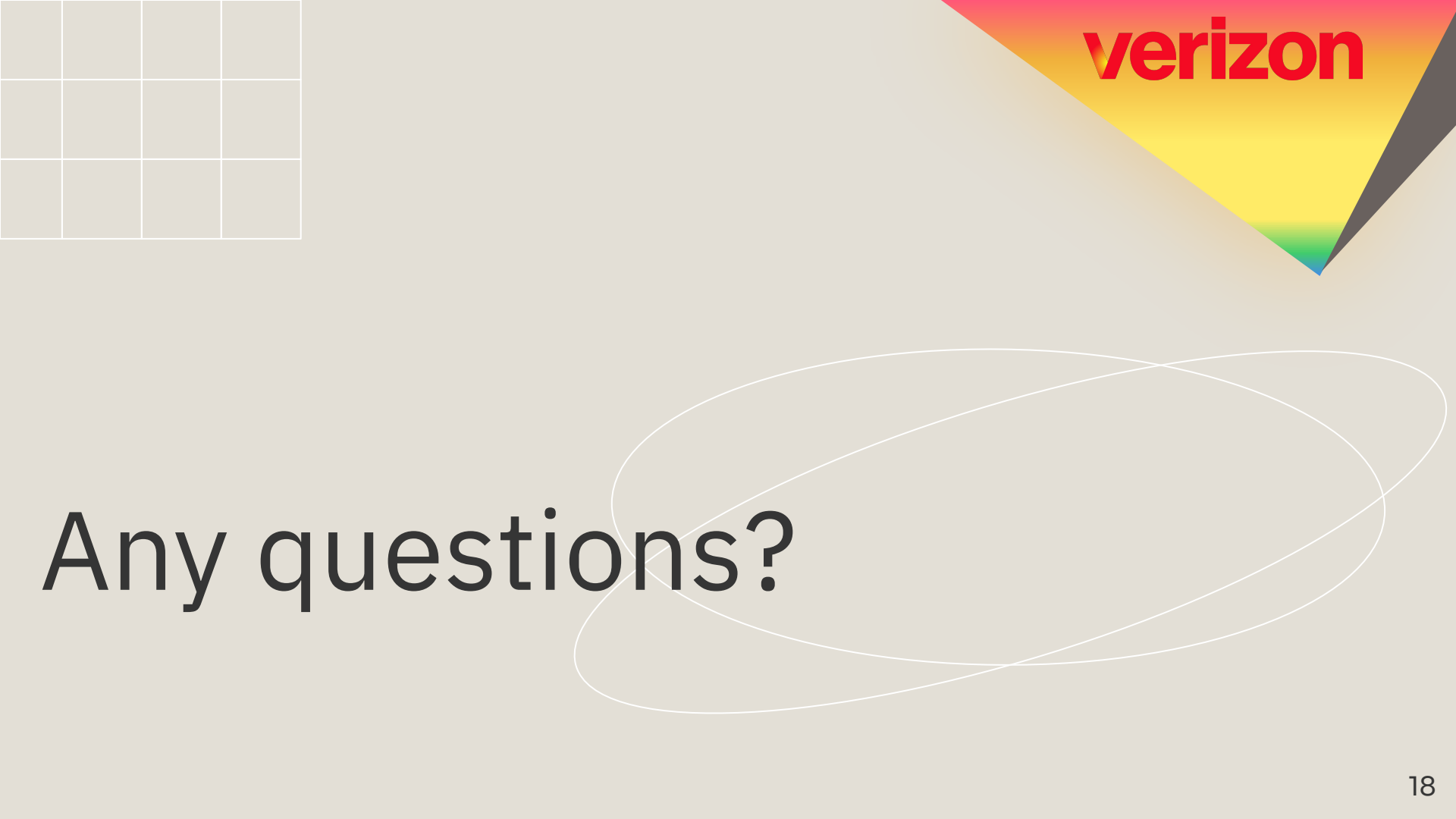
Policy Design &
Decision Strategy
Define Profit-
Maximizing
Threshold
(Approval Policy).

Week 10-12

Pilot Deployment &
Front-End Integration
Front-End Tool
Deployment & User
Testing.

Post-Launch

Performance Monitoring &
Continuous Improvement
Continuous Monitoring &
Threshold Optimization.

The slide features a light grey background. In the top left corner, there is a 4x4 grid of squares. In the bottom right corner, there are two overlapping white ovals. The text "Any questions?" is written in a large, dark grey, sans-serif font, centered horizontally and partially overlapping the right oval.

Any questions?



Thank you

Appendix



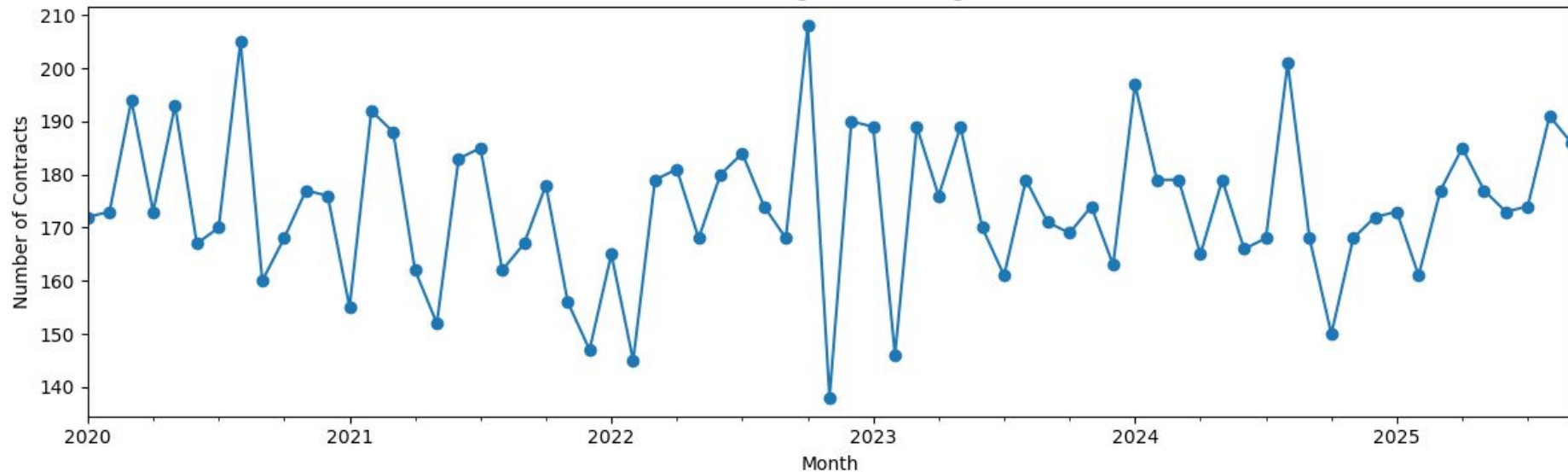
Calculation Inputs

Variable	Source	Value	Notes
ARPA (Q3 2025)	Verizon Earnings	\$147.91	Adjust to the month
Churn (postpaid)	Verizon Earnings	0.91 %	Adjust to the month
LTV (Gross)	ARPA/churn	\$16,259	Per Account
Margin	Reported operating margin	29.4%	Used to derive \$4,780 net LTV
Loss per default	Internal Dataset	\$811	Avg Remaining Balance for defaults only
Default rate	Model test Output	22.9%	vs 24.6% baseline

Appendix

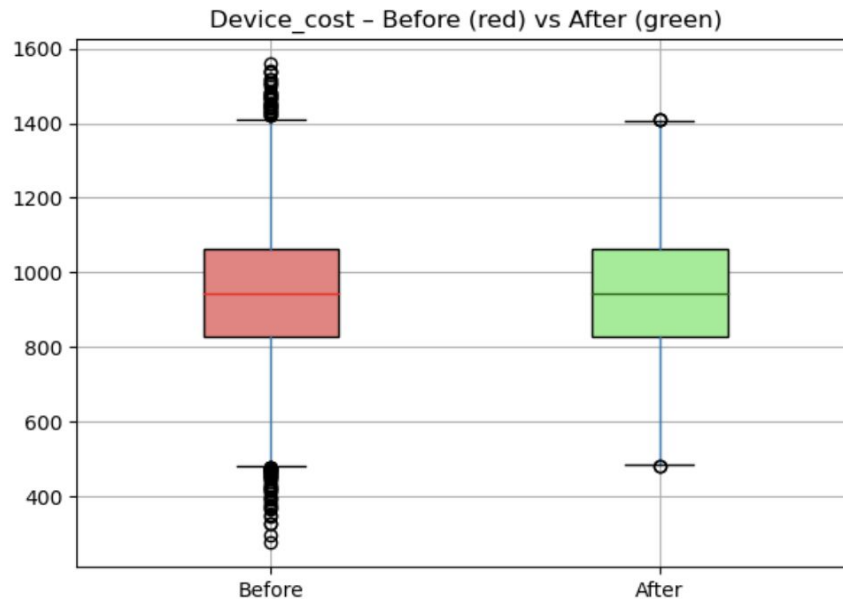
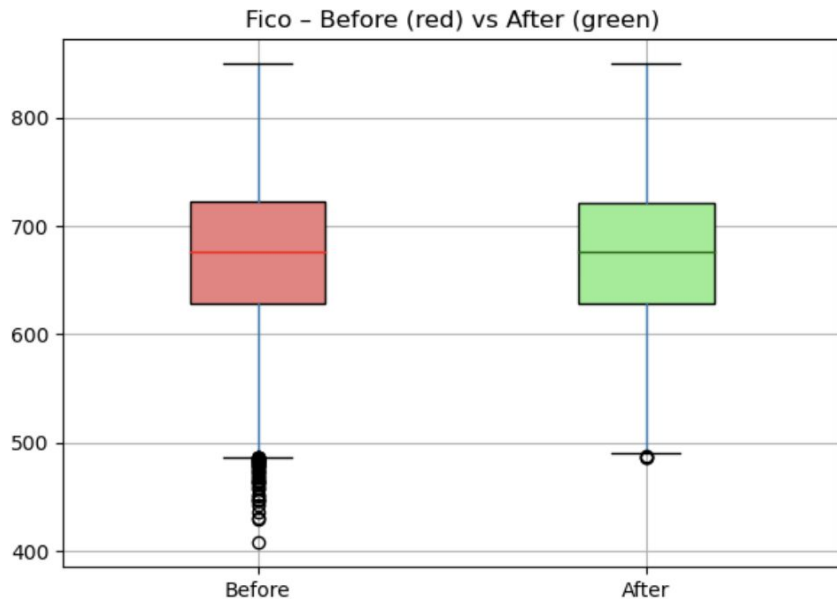


Verizon Device-Financing Contracts Signed Over Time



Appendix

Outlier Removal Impact(IQR Method)



Appendix

```
: baseline_stats = calculate_cm(y_hat, y_test)
baseline_stats
```

```
: {'tp': 0.34722222222221305,
   'fp': 0.06166666666666677,
   'tn': 0.12916666666666698,
   'fn': 0.4619444444444303}
```

```
: my_stats = calculate_cm(y_hat, y_test)
my_stats
```

```
: {'tp': 0.5088740180389943,
   'fp': 0.13034623217922695,
   'tn': 0.099505382601106,
   'fn': 0.26127436718068375}
```